

Annex B

Computational Thinking Exercise: "Smart Vending Machine"

Section: 9-Oxygen

Score: _____

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Scenario

Your school installs a **smart vending machine** to provide snacks and drinks. However, students encounter several issues:

- Sometimes the machine does not give the correct change.
- Items run out, but the machine does not notify anyone.
- Students press the wrong buttons and receive the wrong item.
- The machine is slow when multiple students use it in succession.

Your group's task is to decompose this problem into smaller, manageable parts that could be solved using Computational Thinking (CT) Skills.

Step 1: Identify the Big Problem

Main Problem:

The school's smart vending machine does not work properly and efficiently. This causes problems to the students when errors occur such as incorrect change, out-of-stock items, wrong item selection, and slow service.

Step 2: Identify Three to Four Sub-Problems

Please list possible sub-problems.

1. **Incorrect change** - The students sometimes receive the wrong amount of change After they purchase something from the vending machine.
2. **Out of Stock Notification** - The machine does not notify the staff if it runs out of stock of snacks and drinks. This leaves empty slots on the vending machine.
3. **The Dispense of Wrong Items** - When students accidentally press a wrong button, it cannot be undone, which causes the student to receive the wrong item.
4. **Slow Processing time** - The vending machine takes a long time to process purchases, especially when students use it one after another.

Step 3: Define Computational Thinking Approaches

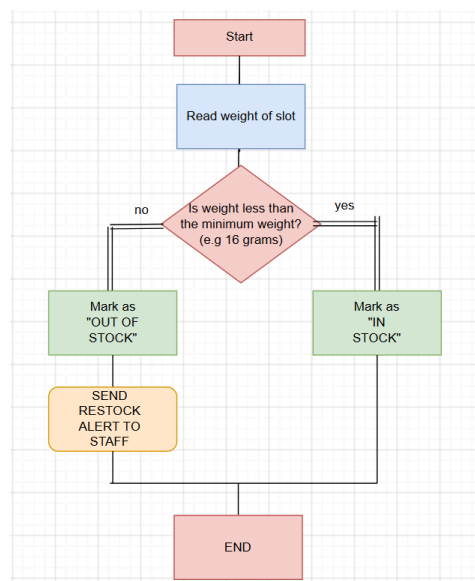
For each sub-problem, identify the appropriate Computational Thinking (CT) skill and suggest a possible solution.

Sub-Problem	CT Skill	Example Solution
Incorrect change	Algorithm	Create an algorithm that accurately calculates the scanned money by the students.
Out of Stock Notification	Pattern Recognition	It monitors inventory levels by the weight of the slots (if it's no longer heavy, it's empty.) This sends a signal to the staff.
The Dispense of Wrong Items	Abstraction	This simplifies the interface of the machine to make it user-friendly. Students can now easily navigate and prevent pressing wrong buttons.
Slow Processing time	Decomposition	Makes the vending machine process simpler tasks one at a time. (payment, item dispensing, and change). This is to optimize each step and make the process faster.

Step 4: Design a Solution

Choose **one** of the identified sub-problems.

Draw a flowchart or write a pseudocode that shows how the problem can be solved.



Flowchart